

Memory Access Integrity Module (MACIM)

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Independent Methodological Authority

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Memory Governance Layer

Governed Space: Memory Access Integrity

Category: AI & Interpretation

Subcategory: Memory Access Governance

Type: Memory Governance Standard Module

Parent Standard: Memory Governance Standard (MGS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 12 June 2026

Compatibility: OOF Methodology OS · Memory Governance Standard (MGS) · Memory Authority Integrity Module (MAUIM) · User Memory Integrity Standard (UMIS) · Agent Memory Integrity Standard (AMIS) · Memory Integrity Standard (MIS) · INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Memory Access Integrity Module (MACIM) defines the structural conditions under which access to memory remains authorized, traceable, accountable, reviewable, governable, and operationally valid throughout the memory lifecycle.

MACIM governs memory access.

The module ensures that memory is accessed only through valid authority, valid permissions, valid governance conditions, and valid operational requirements.

A system satisfies MACIM only if:

- access rights remain identifiable
- access decisions remain traceable
- unauthorized access remains detectable
- access governance remains assessable
- access accountability remains preservable
- memory access remains operationally valid

A memory environment that cannot explain who accessed memory, when memory was accessed, or why memory was accessed does not satisfy MACIM.

Module Operational Role

MACIM defines the access-governance layer of MGS.

Its role is to preserve integrity of memory access operations.

Module Operational Space

MACIM governs:

- memory access
- access permissions
- retrieval authorization
- access accountability
- access governance
- memory visibility
- memory authorization
- access traceability

The module applies wherever memory can be accessed, retrieved, viewed, queried, or utilized.

Module Function

The module applies wherever systems must preserve:

- authorized access
- governance-valid retrieval
- accountable memory utilization
- traceable access activity
- controlled memory visibility
- trustworthy memory operations

Its function is to ensure that memory access remains controlled and governable.

Minimum Implementation Framework

1. Define the Memory Access Object

The organization must define which memory environments require access governance.

This may include:

- AI memory systems
- user memory environments
- organizational knowledge systems
- Human-AI memory environments
- collective memory repositories
- autonomous agent memory
- operational memory systems
- institutional archives

2. Define Memory Access Conditions

The system must define the conditions under which access remains valid.

This includes:

- authorization requirements
- permission requirements
- accountability requirements
- traceability requirements
- oversight requirements
- governance-valid access conditions

3. Define Access Degradation Detection Logic

The system must define how access failures are identified.

This may include:

- unauthorized access
- excessive permissions
- invisible access events
- governance bypass
- access abuse
- permission drift

4. Define Operational Response or Governance Logic

The system must define governance logic for access-integrity failures.

Governance response may include:

- access review
- authorization verification
- governance intervention
- escalation
- permission correction
- access restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- access events
- authorization decisions
- governance reviews
- intervention procedures
- validation activities
- resulting access states

A memory-governance environment must not remain access-valid

if materially significant access activities cannot be reconstructed, reviewed, validated, or governed.

Use Case 1 — Human-AI Memory Environment

Scenario

Humans and AI systems jointly utilize memory resources containing operational knowledge, personal information, and decision-support information.

Application

MACIM governs authorization, visibility, and accountability of memory access.

Result

The environment gains stronger trustworthiness and reduced exposure to unauthorized memory utilization.

Use Case 2 — Enterprise Knowledge Repository

Scenario

An organization maintains a large institutional knowledge repository accessed by employees, managers, contractors, and AI systems.

Application

MACIM governs access permissions, retrieval accountability, and governance oversight.

Result

The organization gains stronger information control and improved governance visibility.

Canonical Closing Statement

Memory Access Integrity Module (MACIM) defines the structural conditions under which access to memory remains authorized, traceable, accountable, reviewable, governable, and operationally valid throughout the memory lifecycle.

**Memory governance cannot remain trustworthy if access becomes uncontrolled.
Memory access integrity therefore becomes a foundational integrity condition
of trustworthy memory governance systems.**

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Canonical Source

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